Mock diagnostic examination - Maths paper

Candidate paper

Time allowed: 90 minutes
Marks for this paper: 55
Marks across both mock papers: 100
Questions: All questions and all parts are compulsory.

Instructions

- Complete this paper independently of the Statistics paper.
- Complete the paper without notes or solutions. You may use the supplied Maths formula sheet.
- A basic non-programmable scientific calculator is permitted, but no question requires one.
- No computational software is needed or permitted. Approved assistive technology, accessibility accommodations and adjusted timing remain permitted where authorised; assistive tools must not perform the assessed mathematics.
- Show the reasoning and intermediate work needed to support each answer. An unsupported answer may not receive full marks.
- Give exact answers unless a question asks for an approximation.
- All vectors are column vectors. Use the course notation throughout.
- For inequality-constrained maximisation, write every inequality as $g_j(\mathbf{x}) \geq 0$ and use the course Kuhn–Tucker convention.

Section M0 — short diagnostic questions (10 marks; 15 minutes)

Question 1 — `MOCK-M0-Q01-PA` (2 marks; 3 minutes) Consider the claim

$$\forall x \in \mathbb{R}, \qquad x^2 \geq x.$$

Write its logical negation using $\exists$, then give one counterexample to the claim.

Question 2 — `MOCK-M0-Q02-PA` (2 marks; 3 minutes) Solve

$$\log(x-1) + \log(x+1) = \log 8$$

over its real domain.

Question 3 — `MOCK-M0-Q03-PA` (2 marks; 3 minutes) Find the first-order Taylor approximation to $f(x) = \sqrt{x}$ about $x = 4$, and use it to approximate $\sqrt{4.4}$.

Question 4 — `MOCK-M0-Q04-PA` (2 marks; 3 minutes) Let $\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^2$ and $h : \mathbb{R}^2 \to \mathbb{R}$ be differentiable. State the dimensions of $\mathbf{J}_{\mathbf{F}}(\mathbf{x})$ and of the column gradient $\nabla h(\mathbf{u})$.

Question 5 — `MOCK-M0-Q05-PA` (2 marks; 3 minutes) Classify the quadratic form

$$q(x, y) = 2x^2 + 2xy + 2y^2$$

by definiteness. Name the test or calculation that supports your conclusion.

Section M1 — linear algebra (15 marks; 25 minutes)

Question 6 — systems, rank and parameter cases (8 marks; 14 minutes)

Part A — `MOCK-M1-Q06-PA` For parameters $a, b \in \mathbb{R}$, consider

$$\begin{aligned} x + y + z &= 1, \\ x + ay + z &= b, \\ x + y + az &= b. \end{aligned}$$

Use Gaussian elimination to classify the system for every pair (a, b) as having a unique solution, no solution or multiple solutions. Give the unique solution when it exists. When $a = 1$, state both the coefficient rank and the augmented-matrix rank for each possible case.

Question 7 — eigenvalues, eigenvectors and diagonalisation (7 marks; 11 minutes)

Part A — `MOCK-M1-Q07-PA` Let

$$\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 0 & 1 \end{bmatrix}.$$

Find the eigenvalues and a basis for each eigenspace. Explain why the chosen eigenvectors form a basis of $\mathbb{R}^2$, then give matrices $\mathbf{P}$ and $\mathbf{D}$ such that $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$.

Section M2 — multivariable calculus (12 marks; 20 minutes)

Question 8 — a vector chain rule (4 marks; 7 minutes)

Part A — `MOCK-M2-Q08-PA` Define $\mathbf{F} : \mathbb{R}^2 \to \mathbb{R}^2$ and $\mathbf{G} : \mathbb{R}^3 \to \mathbb{R}^2$ by

$$\mathbf{F}(u, v) = \begin{bmatrix} u^2 + v \\ uv \end{bmatrix}, \qquad \mathbf{G}(x, y, z) = \begin{bmatrix} x + y \\ yz \end{bmatrix}.$$

Use the multivariable chain rule to calculate $\mathbf{J}_{\mathbf{F} \circ \mathbf{G}}(1, 1, 1)$. Show the dimensions of the two Jacobian matrices before multiplying them.

Question 9 — implicit differentiation (4 marks; 7 minutes)

Part A — `MOCK-M2-Q09-PA` The equation

$$F(x, y) := x^2 + xy + y^2 - 7 = 0$$

holds at $(x, y) = (1, 2)$. State the derivative condition needed for this equation to define y locally as a differentiable function of x. Calculate $\mathrm{d}y/\mathrm{d}x$ at $(1, 2)$, then use it to approximate the change in y when x increases by 0.05.

Question 10 — homogeneity and Euler's theorem (4 marks; 6 minutes)

Part A — `MOCK-M2-Q10-PA` For $x > 0$ and $y > 0$, define

$$p(x, y) = x^{2/5} y^{3/5}.$$

Verify from the definition that p is homogeneous and state its degree. Use Euler's theorem to obtain a relation between p, p_x and p_y, and interpret what happens to p when both inputs are doubled.

Section M3 — curvature and optimisation (18 marks; 30 minutes)

Question 11 — unconstrained optimisation (6 marks; 10 minutes)

Part A — `MOCK-M3-Q11-PA` Let

$$f(x, y) = 4x + 6y - x^2 - xy - y^2, \qquad (x, y) \in \mathbb{R}^2.$$

Find every stationary point and classify it using the Hessian. State the condition that makes your conclusion global on $\mathbb{R}^2$, then give the global optimum value.

Question 12 — equality-constrained optimisation and the envelope theorem (6 marks; 10 minutes)

Part A — `MOCK-M3-Q12-PA` For a resource parameter $a > 0$, consider

$$\max_{x>0,\, y>0} \{2 \log x + \log y\} \quad \text{subject to} \quad 2x + y = a.$$

Encode the constraint as $a - 2x - y = 0$. Form the Lagrangian and solve for $x^*(a)$, $y^*(a)$ and the multiplier $\mu^*(a)$. State a condition that makes the solution the unique global maximiser. Use the envelope theorem to calculate $V'(a)$, where V is the value function, and interpret the sign of the multiplier under the stated constraint encoding.

Question 13 — inequality constraints and a binding resource limit (6 marks; 10 minutes)

Part A — `MOCK-M3-Q13-PA` Consider

$$\max_{x,y} \left\{ 3x + 2y - \frac{1}{2}x^2 - y^2 \right\}$$

subject to

$$x \geq 0, \qquad y \geq 0, \qquad 2 - x - y \geq 0.$$

Form the Lagrangian and write the complete Kuhn–Tucker conditions under the course convention. Solve the problem, reporting all multipliers and identifying every active and slack constraint. Use curvature and the feasible set to justify whether the solution is the unique global maximiser.

Formula sheet

This sheet accompanies the 90-minute, 55-mark Maths mock paper.

General conventions

- Give exact answers unless a question asks for an approximation.
- Scalars use italic lower-case letters. Vectors are bold lower-case column vectors, and matrices are bold upper-case letters.
- Use $\mathbf{A}^\mathsf{T}$, $\det(\mathbf{A})$, $\operatorname{rank}(\mathbf{A})$ and $[\mathbf{A} \mid \mathbf{b}]$ for transpose, determinant, rank and an augmented matrix.
- Gradients are column vectors. For $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m$, use $\mathbf{J}_\mathbf{F}$ for its Jacobian.
- For a maximisation problem with inequality constraints, write each inequality as $g_j(\mathbf{x}) \geq 0$ and use the course Kuhn–Tucker convention.

Formula provision

No additional formulae are supplied. The derivative rules, Taylor formula, matrix tests, chain rule, implicit-differentiation rule, Euler relation, optimisation conditions and Kuhn–Tucker conditions used in this paper are part of the assessed material.